MSc QFAS Thesis Proposal - Internship at Cardano

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Optimization and hedging of liability-driven investment portfolios under the Wet toekomst pensioenen: earning the protection return under market frictions

1 Company description

Cardano is specialized in risk - and asset management solutions for institutional investors with a long-term investment horizon, in particular pension funds. Cardano designs risk management strategies for liability-driven investment portfolios (LDI), besides offering public and private market solutions for sustainable (e.g. ESG/Impact) investing. This thesis will be written within Cardano Risk Management, and my company supervisor is Roel Mehlkopf, Pension Fund Advisor within the Client Advisory team (roel.mehlkopf@cardano.com).

2 Background and Research Question

Cardano advises pension funds on the management of their LDI portfolios, which are exposed to a range of risks, e.g. inflation risk, currency risk, longevity risk and credit risk, with the dominant one being interest rate risk. Pension entitlements are long-dated and valued at market rates, hence a decline in the term structure of interest rates raises the present value of future benefit payments. To manage this exposure, Cardano's portfolio management department constructs tailor-made derivative overlay strategies that gain value when rates decline. Each overlay strategy has different instrument mixes which evolve over time as market conditions and client mandates change.

The transition from the *defined benefit* pension system under the Financieel Toetsingskader (FTK) to the new *defined contribution* system under the Wet toekomst pensioenen (Wtp) changes the purpose and composition of this overlay. Specifically, under the *solidaire premieregeling* every participant holds a personal pension capital which is collectively invested. The resulting return from this collective investment can be decomposed into a compensation for the passage of time, the *protection return* (beschermingsrendement) which hedges interest rate risk, and the *residual return* (overrendement) obtained from - in general - risky assets. This mechanism is designed to protect the older participants against interest rate fluctuations, while younger participants can benefit from taking extra risk. The protection return is thus the return allocated to an age cohort in order to neutralize the effect of a change in the term structure on the value of entitlement. It can be determined indirectly via the term structure of interest rates (*rentetermijnstructuur, RTS*) published by the Dutch Central Bank, or directly using a segregated hedging portfolio (*matchingportefeuille).* Potter van Loon & Alserda (2025) compare both methods and find that the allocated protection return can deviate materially from what the underlying portfolio delivers.

The difference between the protection return a cohort requires and what the overlay actually delivers, i.e. the *mismatch return*, is ultimately borne by participants. The method through which the theoretical protection return is determined governs where this difference is recorded: under the indirect method it flows into

the residual return and is borne collectively, but mostly by younger cohorts. Under the direct method the allocated returns equals realized return by construction. However, the mismatch is a measurable economic quantity independent of accounting terminology. Hence, this thesis uses the indirect method as baseline, since it ensures separate observability of protection and overlay return. The mismatch arises from market frictions which the theoretical construction ignores: beyond the yield curve point of 30 years the RTS is extrapolated towards an *ultimate forward rate*, hence the very long end of the discount curve cannot be replicated by traded instruments - this affects cohorts with the longest entitlements; swaps are unfunded and leave capital available for other investments but consume collateral, whereas bonds are funded and earn a spread over the discount curve, so which instrument is cheaper depends on the swap spread (which itself depends on pension hedging demand Klingler & Sundaresan (2019)); liquidity held against margin obligations is subject to cash drag, and in case of convex overlay instruments the hedge ratio drifts from its target without any trading, so that correcting it costs transaction fees and collateral. These frictions are currently relevant, as Jansen et al. (2026) recently found that margin calls for Dutch pension funds exceed available cash buffers for 62% of evaluated funds. All in all, the following research question can be formulated, including various sub-questions.

How should a Dutch pension fund construct and manage its interest rate overlay so as to deliver the theoretical protection return required under Wtp at the lowest risk of mismatch, an what does the efficient frontier of this LDI problem look like?

- **Instrument choice**: What mix of instruments - e.g. cash, bonds, interest rate swaps, swaptions - delivers the required protection at the lowest mismatch risk, and how does that mix depend on market conditions, e.g. the swap spread?
- **Collateral and cash drag**: How much liquidity has to be held to meet margin calls on the overlay, and what does holding it cost in terms of total portfolio or protection return?
- **Dynamic management**: How is the mismatch frontier affected by hedge-ratio bandwidth rules, and what is the effect of explicitly accounting for transaction and collateral costs?
- **Robustness**: Is the optimal overlay stable across interest rate regimes, and is it feasible in case of a liquidity stress event?

3 Expected Results

The main deliverable is an efficient frontier for LDI portfolio optimization and overlay construction under the Wtp. The frontier consists out of the attainable combinations of expected mismatch return and a risk measure of that mismatch return under hedge and liquidity constraints. Every point on the frontier is a portfolio of overlay instruments, hence it can be used for decision making in interest rate hedging. Moreover, a decomposition of the mismatch return into its drivers, liquidity buffer insights, and an assessment of how much the frontier moves in various market environments can be obtained; contributing to overlay strategy development. In case of dynamic rebalancing, a point on the frontier represents a rebalancing policy rather than a deterministic instrument mix, hence a frontier/chart can be constructed per policy by varying the parameter which makes its trading decisions, e.g. the hedge ratio bandwidth. Since a static mix is a policy which never trades, the distance between the static and the dynamic frontier could indicate the value of rebalancing.

4 Methodology and Data

The methodology stated below serves as a tentative plan of action and is subject to adjustments; a literature study will precede it. Note that the proposed frontier is derived by simulation, hence the first three steps build this simulation engine, the fourth turns it into an optimization problem and the fifth yields an extension.

1. **Laying the foundation**: Construct a representative LDI portfolio, thus pension entitlements with a cohort structure, and value these on the DNB RTS. Subsequently, derive the required protection profile of the entitlements accounting for protection level (i.e. *toedeling* or hedge ratio)

2. **Build market environment**: Select a (or more) affine interest rate models and calibrate them to the current EUR curve for simulation purposes. The affine property ensures (near) closed-form bond and swaption pricing, which matters since a large number of instrument valuations is required. Extend the model with a stochastic swap spread, such that attractiveness of bonds and swaps differs among simulations.

3. **Simulate the overlay**: For a candidate overlay instrument mix, simulate the portfolio along generated paths and track the realized overlay return taking transaction costs, collateral dynamics, and liquidity buffer into account. Comparing this with the theoretical protection return computed from the simulated RTS on the same path yields the mismatch return, hence a distribution of mismatch returns is obtained per overlay instrument mix.

4. **Construct the frontier**: A "mean-variance" type of problem is obtained with different axes, which are still subject to discussion at this stage. The choice variable is the instrument configuration, the return axis is the mismatch return, and the risk axis can be a risk measure of that mismatch (e.g. VaR, CVaR) together with a measure of collateral/liquidity. The required protection profile enters as a constraint, e.g. hedge ratio bandwidth, which can represent various risk levels.

5. **Extension for dynamic rebalancing**: The setup can be extended by allowing a rule-based protocol/policy to trade along each path. This allows decision making through dynamic rebalancing while interpreting market info and adhering to constraints, and yields insight into how an overlay strategy should be developed under the new pension contract.

Data: Market data is available through the Bloomberg Terminal. For example the EUR swap curve, government bond curves, repo and money market rates. The historical DNB RTS is published directly. Representative cohort composition and realistic collateral dynamics are obtained from Cardano. All remaining data, i.e. the scenario paths on which the optimization and the evaluation are run, can be simulated.

Relevant literature

Gârleanu, N. & Pedersen, L. H. (2013), 'Dynamic trading with predictable returns and transaction costs', *The Journal of Finance* **68**(6), 2309–2340.

Jansen, K. A. E., Klingler, S., Ranaldo, A. & Duijm, P. (2026), 'Pension liquidity risk', *The Review of Financial Studies* . Advance access publication July 27, 2026, https://doi.org/10.1093/rfs/hhag068.

Klingler, S. & Sundaresan, S. (2019), 'An explanation of negative swap spreads: Demand for duration from underfunded pension plans', *The Journal of Finance* **74**(2), 675–710.

Potter van Loon, R. & Alserda, G. (2025), 'Beschermingsrendement: Theoretisch of praktisch?', *VBA Journaal* (160), 30–36.

Rockafellar, R. T. & Uryasev, S. (2000), 'Optimization of conditional value-at-risk', *The Journal of Risk* **2**(3), 21–41.

Sharpe, W. F. & Tint, L. G. (1990), 'Liabilities - a new approach', *The Journal of Portfolio Management* **16**(2), 5–10.